

WEEK 5 ASSIGNMENT

POWER BI – SALES DATA ANALYSIS AND DASHBOARD

Introduction

Power BI is a business intelligence and data visualization tool developed by Microsoft. It is used to import, clean, transform, model and visualize data in an interactive manner.

In this assignment, a sales dataset consisting of information about customers, products, sales orders, resellers, sales territories and dates was analyzed using Power BI. The data was prepared and connected using relationships, followed by the creation of DAX measures and interactive visualizations.

The main objective of this assignment is to analyze sales performance and develop an interactive dashboard that provides meaningful business insights.

TASK 1: POWER BI INTRODUCTION AND DATA IMPORT

Objective

The objective of this task is to understand the Power BI environment and import the required datasets into Power BI for further analysis.

Description

Power BI provides different components such as Power Query, Data View, Model View and Report View. These components help in preparing data, establishing relationships and creating interactive reports.

For this project, multiple datasets related to sales were imported into Power BI. The imported tables include:

- Customer_data
- Date_data
- Product_data
- Reseller_data
- Sales Order_data
- Sales Territory_data
- Sales_data

The **Sales_data** table contains important numerical information such as Sales Amount, Order Quantity, Product Standard Cost, Unit Price and Total Product Cost. Other tables provide additional information about customers, products, dates, resellers and territories.

Procedure

1. Opened Microsoft Power BI Desktop.
2. Selected **Get Data**.
3. Imported the required sales-related datasets.
4. Loaded the tables into Power BI.
5. Verified the imported columns and data types.

6. Opened the Model View to examine the relationships between the tables.

Result

All required datasets were successfully imported into Power BI and were available for data cleaning, modeling and visualization.

SalesOrderLineKey	ResellerKey	CustomerKey	ProductKey	OrderDateKey	DueDateKey	ShipDateKey	SalesTerritoryKey	Order Quantity	Unit Price	Extended Amount
43659001	676	-1	349	20170702	20170712	20170709	2024.994	1	2024.994	2024.99
43659002	676	-1	350	20170702	20170712	20170709	2024.994	5	3	6074.98
43659003	676	-1	351	20170702	20170712	20170709	2024.994	5	1	2024.99
43659004	676	-1	344	20170702	20170712	20170709	2039.994	5	1	2039.99
43659005	676	-1	345	20170702	20170712	20170709	2039.994	5	1	2039.99
43659006	676	-1	346	20170702	20170712	20170709	2039.994	5	2	4079.98
43659007	676	-1	347	20170702	20170712	20170709	2039.994	5	1	2039.99
43659008	676	-1	229	20170702	20170712	20170709	28.8404	5	3	86.521
43659009	676	-1	235	20170702	20170712	20170709	28.8404	5	1	28.8404
43659010	676	-1	238	20170702	20170712	20170709	5.7	5	6	34
43659011	676	-1	223	20170702	20170712	20170709	5.1865	5	2	10.37
43659012	676	-1	220	20170702	20170712	20170709	20.1865	5	4	80.74
43660001	117	-1	326	20170705	20170715	20170712	419.4589	5	1	419.4589
43660002	117	-1	319	20170705	20170715	20170712	874.794	5	1	874.794
43661001	442	-1	300	20170705	20170715	20170712	809.76	6	1	809.76
43661002	442	-1	296	20170705	20170715	20170712	714.7043	6	1	714.7043
43661003	442	-1	304	20170705	20170715	20170712	714.7043	6	2	1429.408
43661004	442	-1	223	20170705	20170715	20170712	5.1865	6	4	20.74
43661005	442	-1	232	20170705	20170715	20170712	28.8404	6	4	115.361
43661006	442	-1	293	20170705	20170715	20170712	722.5949	6	2	1445.185
43661007	442	-1	348	20170705	20170715	20170712	2024.994	6	3	6074.98
43661008	442	-1	351	20170705	20170715	20170712	2024.994	6	2	4049.98
43661009	442	-1	220	20170705	20170715	20170712	20.1865	6	2	40.37
43661010	442	-1	292	20170705	20170715	20170712	818.7	6	2	1637
43661011	442	-1	349	20170705	20170715	20170712	2024.994	6	4	8099.91
43661012	442	-1	346	20170705	20170715	20170712	2039.994	6	2	4079.98
43661013	442	-1	235	20170705	20170715	20170712	28.8404	6	2	57.6808
43661014	442	-1	350	20170705	20170715	20170712	2024.994	6	2	4049.98

1) Power BI Data/Model view showing the imported tables.

TASK 2: DATA CLEANING AND PREPARATION

Objective

The objective of this task is to clean and prepare the imported data so that it can be used accurately for analysis and visualization.

Description

Data cleaning is an important step in data analysis because incorrect data types, missing values, duplicate records and inconsistent values can affect the accuracy of the analysis.

The imported datasets were examined using Power BI's data preparation features. The columns were checked to ensure that numerical fields were suitable for calculations and categorical fields were suitable for filtering and grouping.

The sales data contains important fields such as:

- CustomerKey
- ProductKey
- ResellerKey
- SalesTerritoryKey
- OrderDateKey

- ShipDateKey
- SalesAmount
- OrderQuantity
- UnitPrice
- TotalProductCost
- ProductStandardCost

Data Preparation Performed

The following preparation activities were carried out:

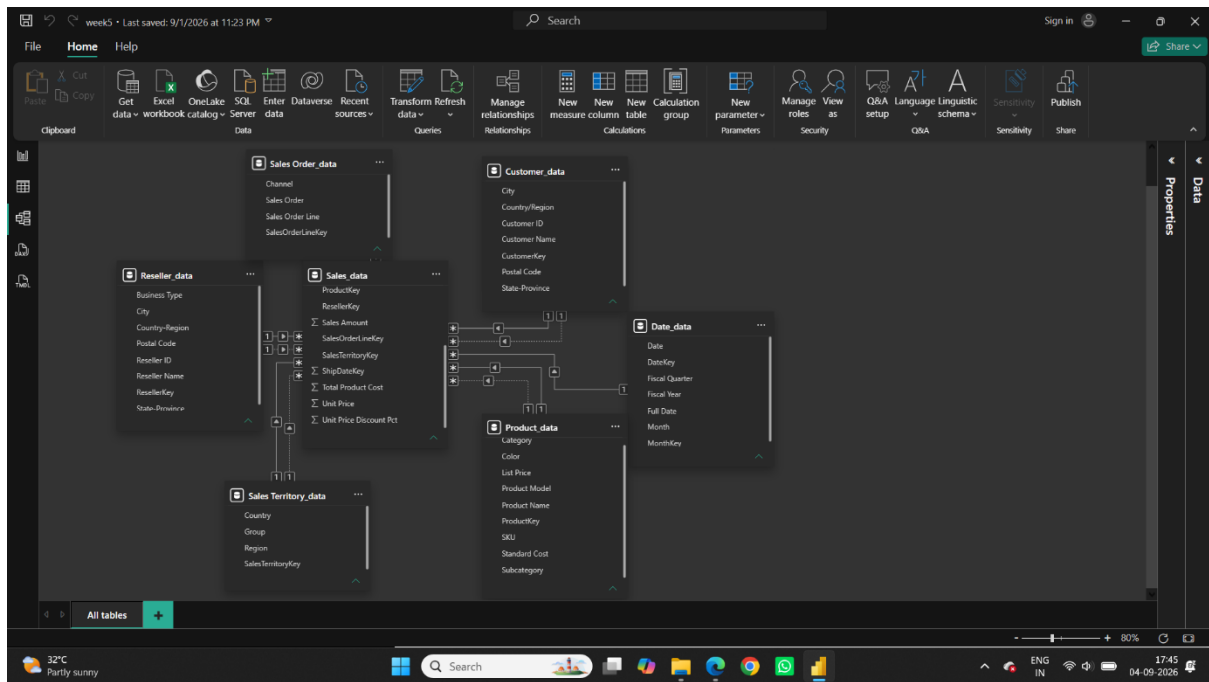
1. Examined the structure of each imported table.
2. Checked the column names and data types.
3. Identified numerical fields required for calculations.
4. Identified categorical fields required for visualization.
5. Verified the key columns used for establishing relationships.
6. Checked the data for consistency.
7. Prepared the tables for data modeling and visualization.

Result

The datasets were successfully prepared for further analysis. The cleaned and structured data was then used to create relationships and DAX calculations.

Table.TransformColumnTypes(*Promoted Headers*, {"SalesOrderLineKey", Int64.Type}, {"ResellerKey", Int64.Type}, {"CustomerKey", Int64.Type}, {"ProductKey", Int64.Type}, {"OrderDateKey", Date.Type}, {"DueDateKey", Date.Type}, {"ShipDateKey", Date.Type})

	SalesOrderLineKey	ResellerKey	CustomerKey	ProductKey	OrderDateKey	DueDateKey	ShipDateKey
1	43659001	676	-1	349	20170702	20170712	
2	43659002	676	-1	350	20170702	20170712	
3	43659003	676	-1	351	20170702	20170712	
4	43659004	676	-1	344	20170702	20170712	
5	43659005	676	-1	345	20170702	20170712	
6	43659006	676	-1	346	20170702	20170712	
7	43659007	676	-1	347	20170702	20170712	
8	43659008	676	-1	229	20170702	20170712	
9	43659009	676	-1	235	20170702	20170712	
10	43659010	676	-1	218	20170702	20170712	
11	43659011	676	-1	223	20170702	20170712	
12	43659012	676	-1	220	20170702	20170712	
13	43660001	117	-1	326	20170705	20170715	
14	43660002	117	-1	319	20170705	20170715	
15	43661001	442	-1	300	20170705	20170715	
16	43661002	442	-1	296	20170705	20170715	
17	43661003	442	-1	304	20170705	20170715	
18	43661004	442	-1	223	20170705	20170715	
19	43661005	442	-1	232	20170705	20170715	
20	43661006	442	-1	293	20170705	20170715	
21	43661007	442	-1	348	20170705	20170715	
22	43661008	442	-1	351	20170705	20170715	
23	43661009	442	-1	220	20170705	20170715	
24	43661010	442	-1	292	20170705	20170715	
25	43661011	442	-1	349	20170705	20170715	
26	43661012	442	-1	346	20170705	20170715	



2) Power Query/Data View showing the prepared data.

TASK 3: DATA MODELING AND RELATIONSHIPS

Objective

The objective of this task is to create a suitable data model by establishing relationships between the different tables.

Description

Data modeling connects related tables using common key fields. A proper data model allows information from different tables to work together when creating reports and visualizations.

The **Sales_data** table acts as the main fact table because it contains transaction-level sales information. The other tables provide descriptive information about customers, products, dates, resellers and sales territories.

The following relationships were established:

- Customer_data → Sales_data using **CustomerKey**
- Product_data → Sales_data using **ProductKey**
- Reseller_data → Sales_data using **ResellerKey**
- Sales Territory_data → Sales_data using **SalesTerritoryKey**
- Date_data → Sales_data using the appropriate date key
- Sales Order_data → Sales_data using the relevant sales order key

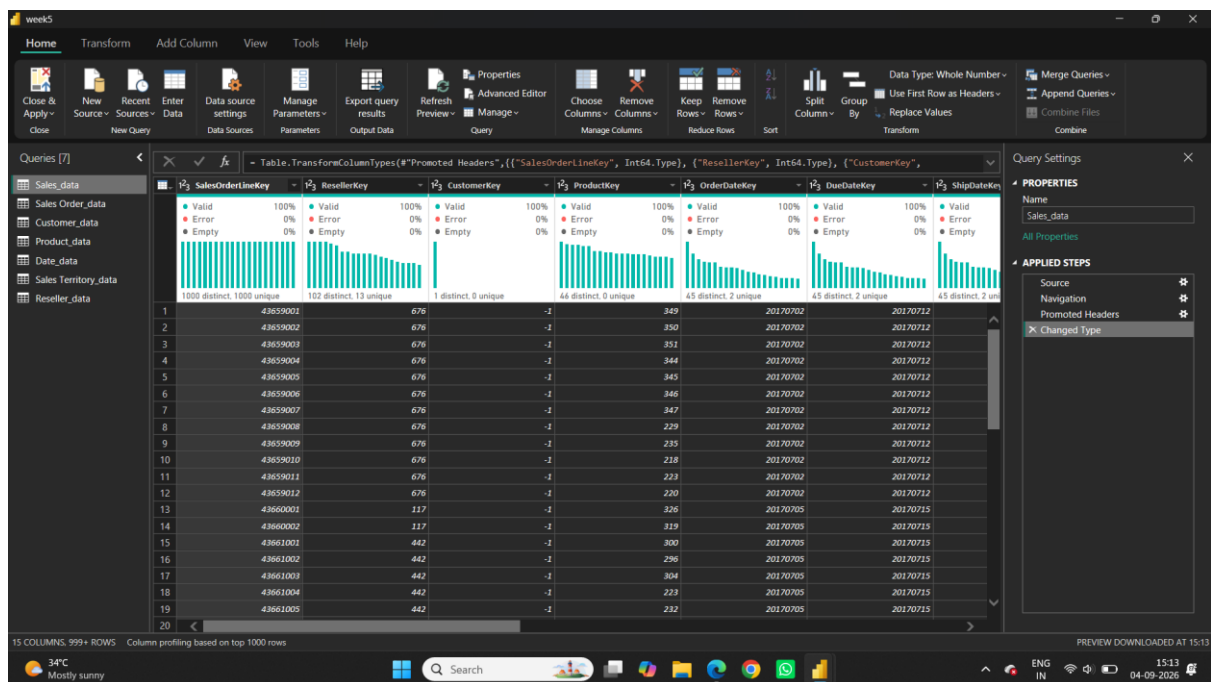
Procedure

1. Opened **Model View** in Power BI.

2. Examined the available tables.
3. Identified the common key columns.
4. Created relationships between the related tables.
5. Checked the relationship directions and cardinality.
6. Verified that the model allowed data to flow correctly between dimension tables and the Sales_data table.

Result

A relational data model was successfully created. The model allows sales information to be analyzed based on products, customers, dates, territories and other dimensions.



3) Your Model View screenshot showing all relationships.

TASK 4: DAX CALCULATED MEASURES

Objective

The objective of this task is to create DAX measures for calculating important sales performance indicators.

Description

DAX (Data Analysis Expressions) is the formula language used in Power BI for creating calculated columns, calculated tables and measures.

In this task, five measures were created to summarize the sales data and display important Key Performance Indicators (KPIs).

The measures were stored under **Sales_data** because the calculations are based on sales information.

1. Total Sales

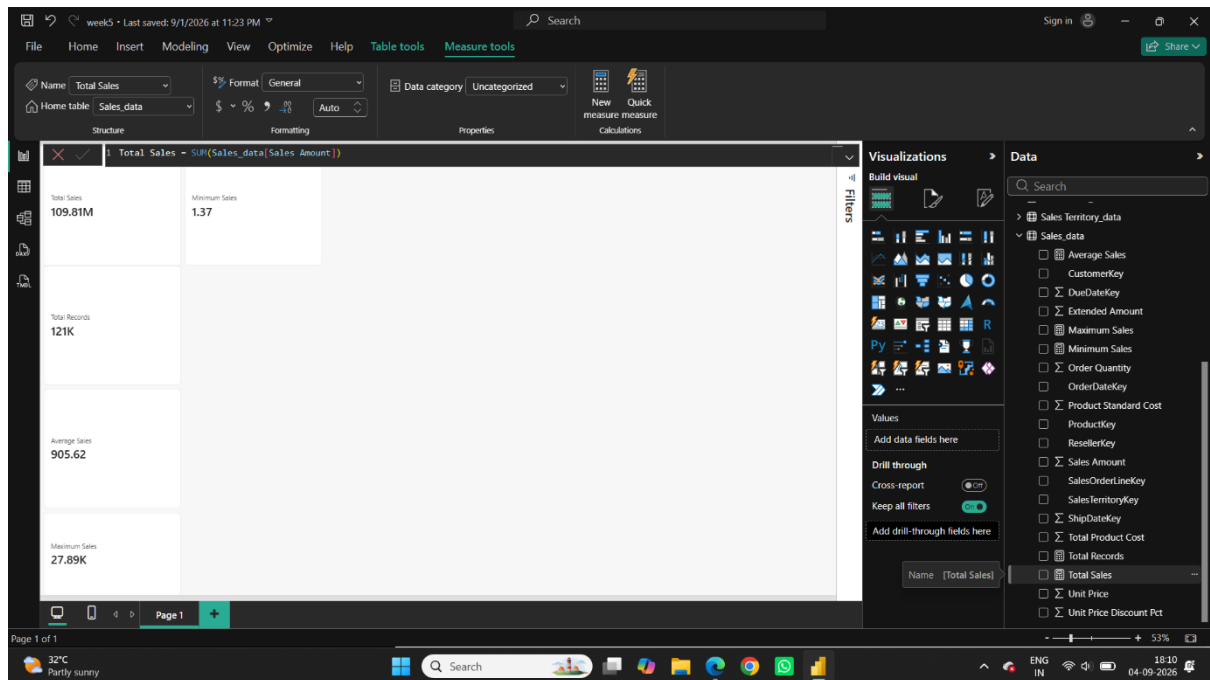
DAX Formula

Total Sales = SUM(Sales_data[Sales Amount])

Description

This measure calculates the total sales generated from all sales transactions.

Output



The dashboard shows total sales of approximately:

₹109.81 Million

2. Average Sales

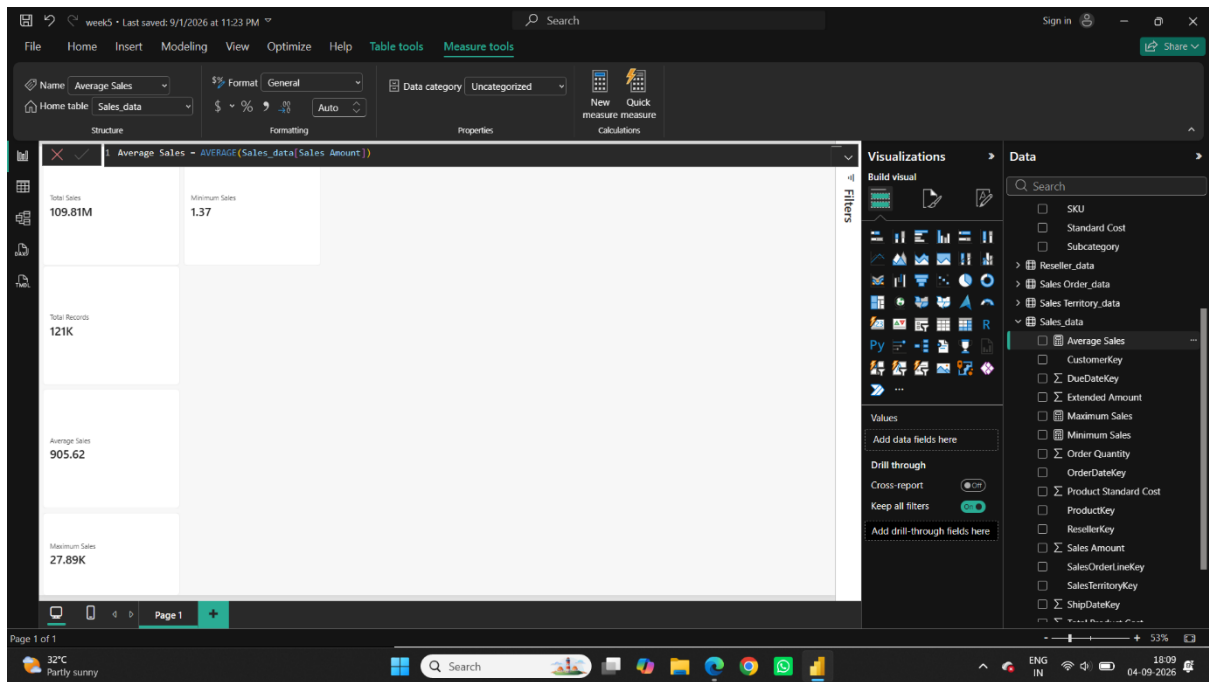
DAX Formula

Average Sales = AVERAGE(Sales_data[Sales Amount])

Description

This measure calculates the average sales value across the sales records.

Output



The dashboard shows an average sales value of approximately:

₹905.62

3. Total Records

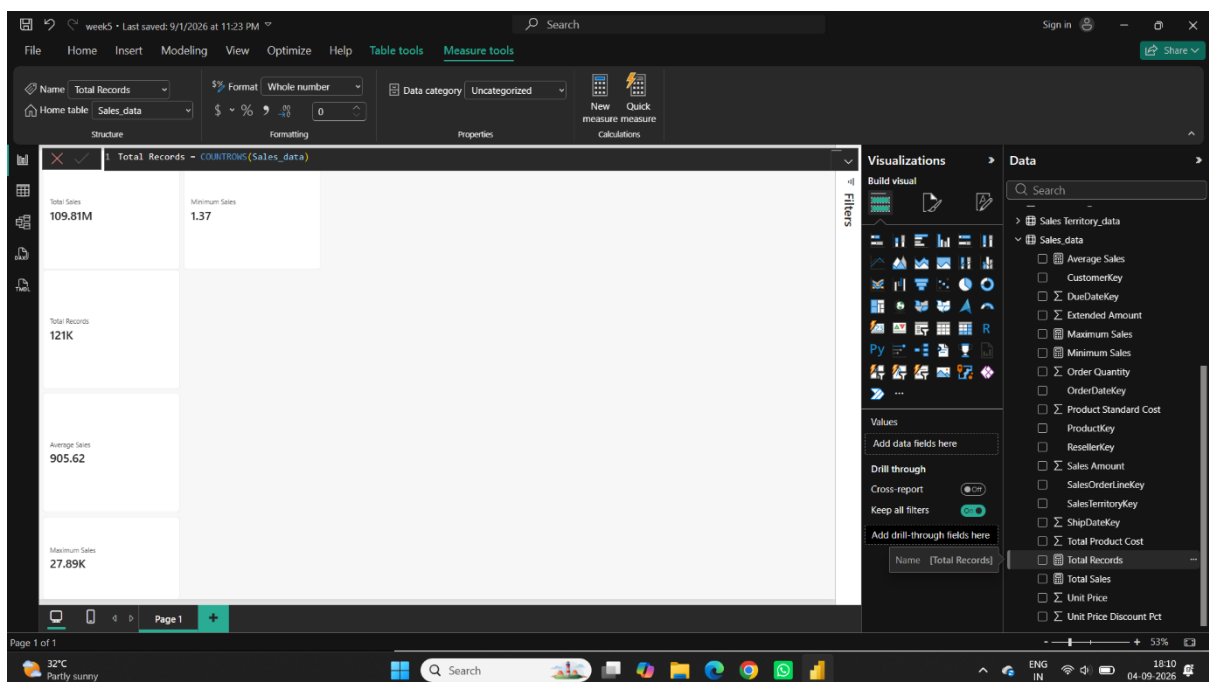
DAX Formula

Total Records = COUNTROWS(Sales_data)

Description

This measure counts the total number of records available in the Sales_data table.

Output



The dashboard displays approximately:

121K records

4. Maximum Sales

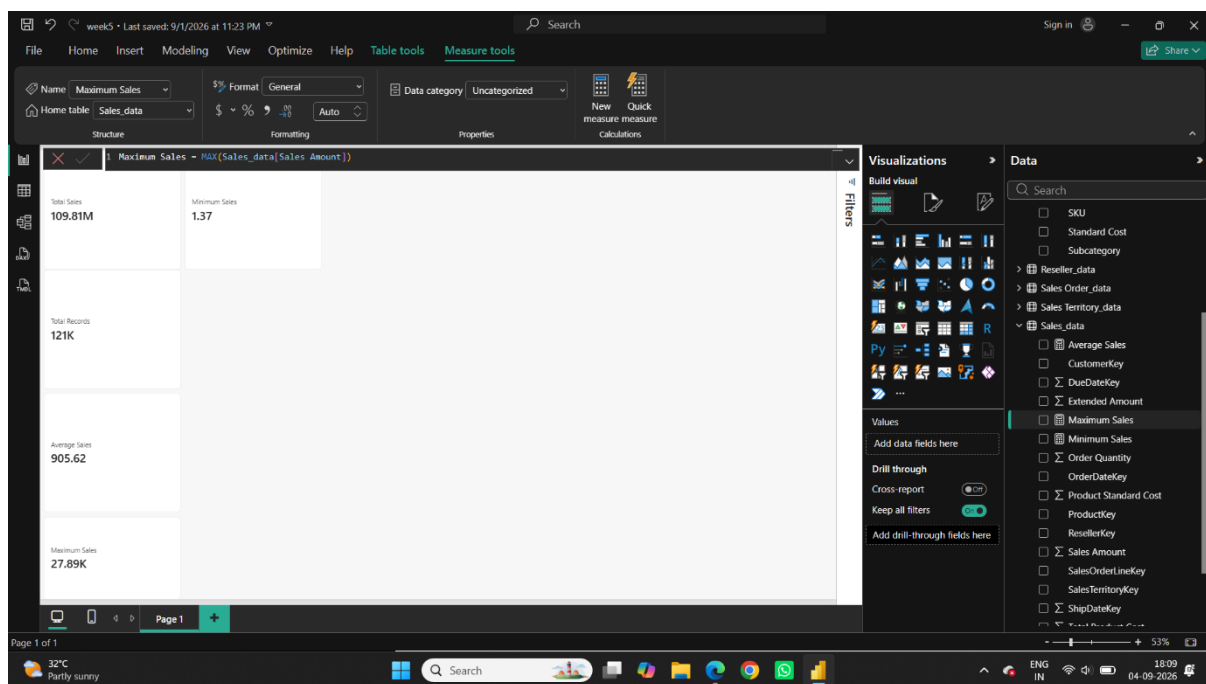
DAX Formula

Maximum Sales = MAX(Sales_data[Sales Amount])

Description

This measure identifies the highest individual sales amount recorded in the dataset.

Output



The maximum sales value displayed on the dashboard is approximately:

₹27.89K

5. Minimum Sales

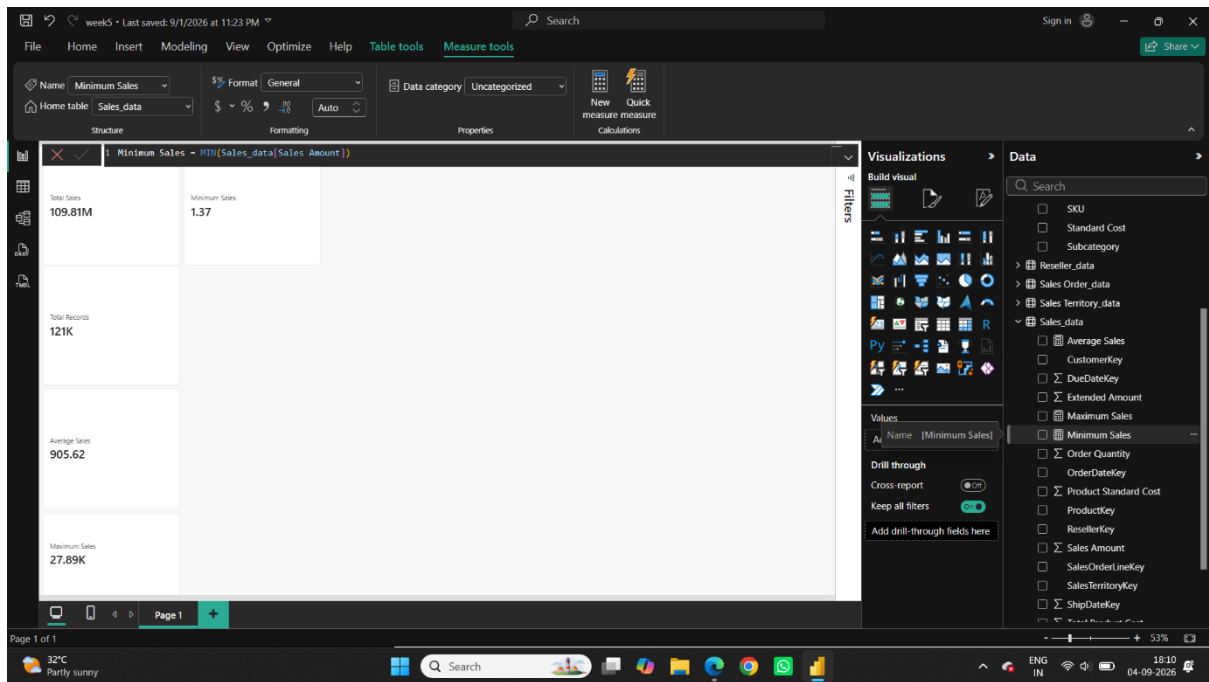
DAX Formula

Minimum Sales = MIN(Sales_data[Sales Amount])

Description

This measure identifies the lowest individual sales amount recorded in the dataset.

Output

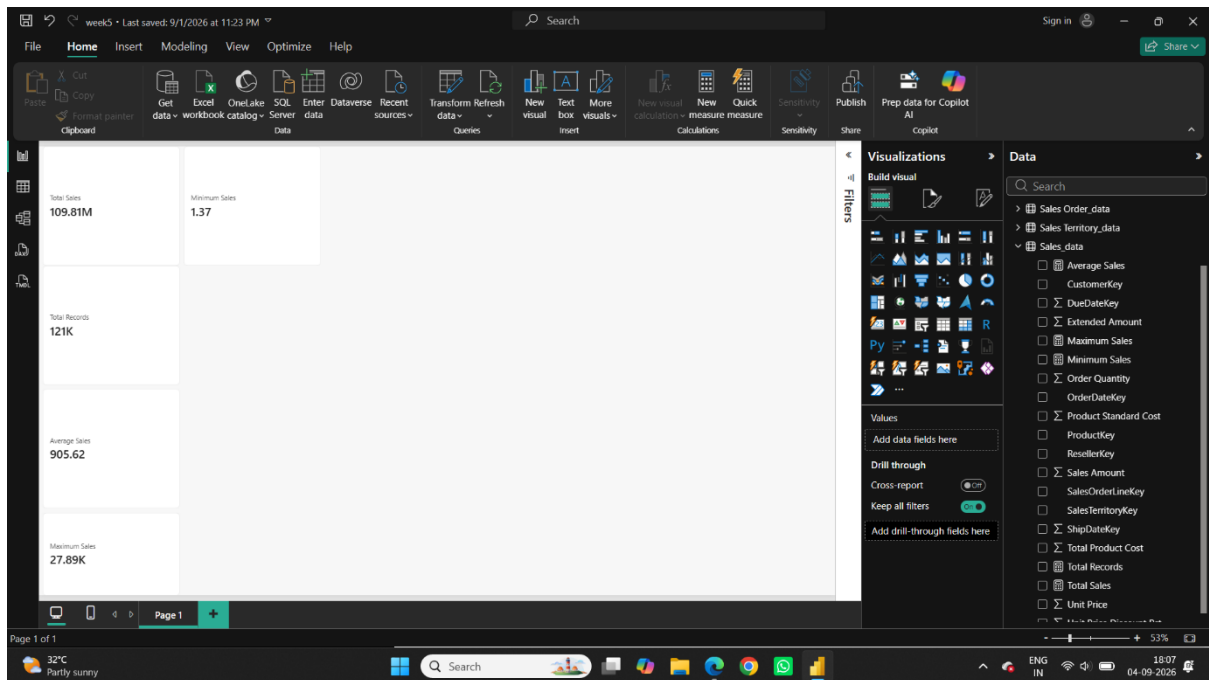


The minimum sales value displayed on the dashboard is approximately:

₹1.37

Result:

Five DAX measures were successfully created and used as KPI cards in the Power BI dashboard.



4) KPI cards

TASK 5: POWER BI VISUALIZATIONS

Objective

The objective of this task is to create meaningful visualizations that represent sales performance and help identify important patterns in the data.

Description

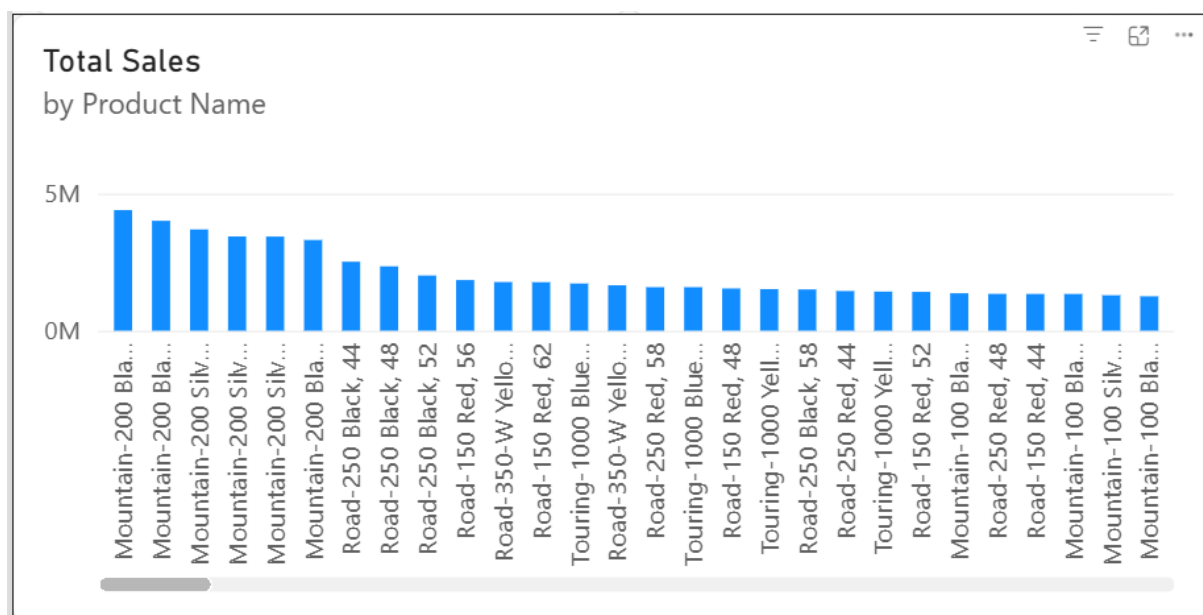
Different Power BI visualizations were created using the prepared sales data and DAX measures.

The dashboard contains the following visualizations.

1. Total Sales by Product Name

A bar/column chart was created to compare the total sales generated by different products.

This visualization helps identify products that contribute significantly to the company's revenue.



5) Total Sales by Product Name chart

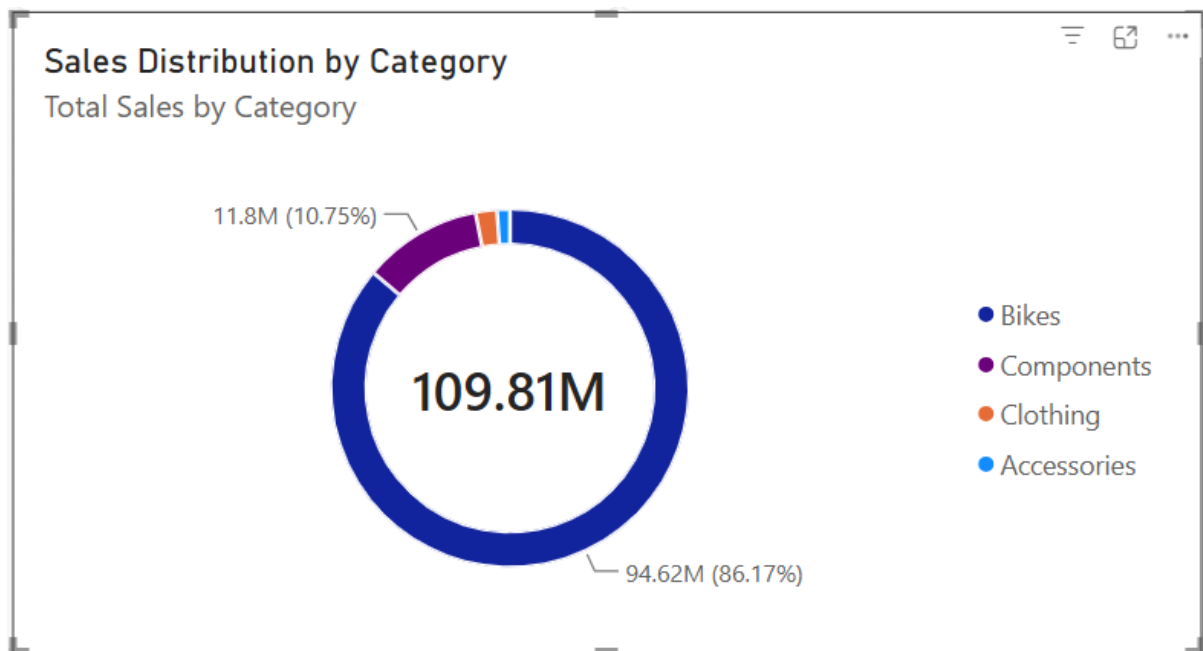
2. Sales Distribution by Category

A doughnut chart was created to show how total sales are distributed among different product categories.

The categories include:

- Bikes
- Components
- Clothing
- Accessories

The visualization clearly shows that **Bikes contribute the largest share of total sales.**



6) Sales Distribution by Category doughnut chart

3. Total Sales by Product Category

A chart was created to compare sales across product categories.

The visualization shows that the Bikes category has considerably higher sales compared with Components, Clothing and Accessories.



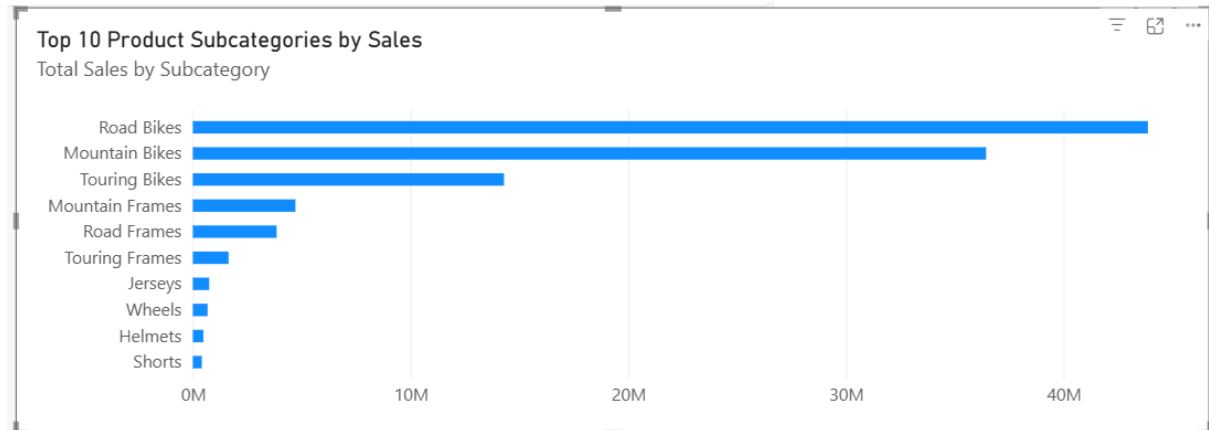
7) Total Sales by Product Category chart

4. Top 10 Product Subcategories by Sales

A horizontal bar chart was created to display the top 10 product subcategories based on sales.

The visualization shows that **Road Bikes** and **Mountain Bikes** are among the strongest-performing subcategories.

This helps the business identify which product groups generate the most revenue.



8) Top 10 Product Subcategories by Sales

5. Product Sales Table

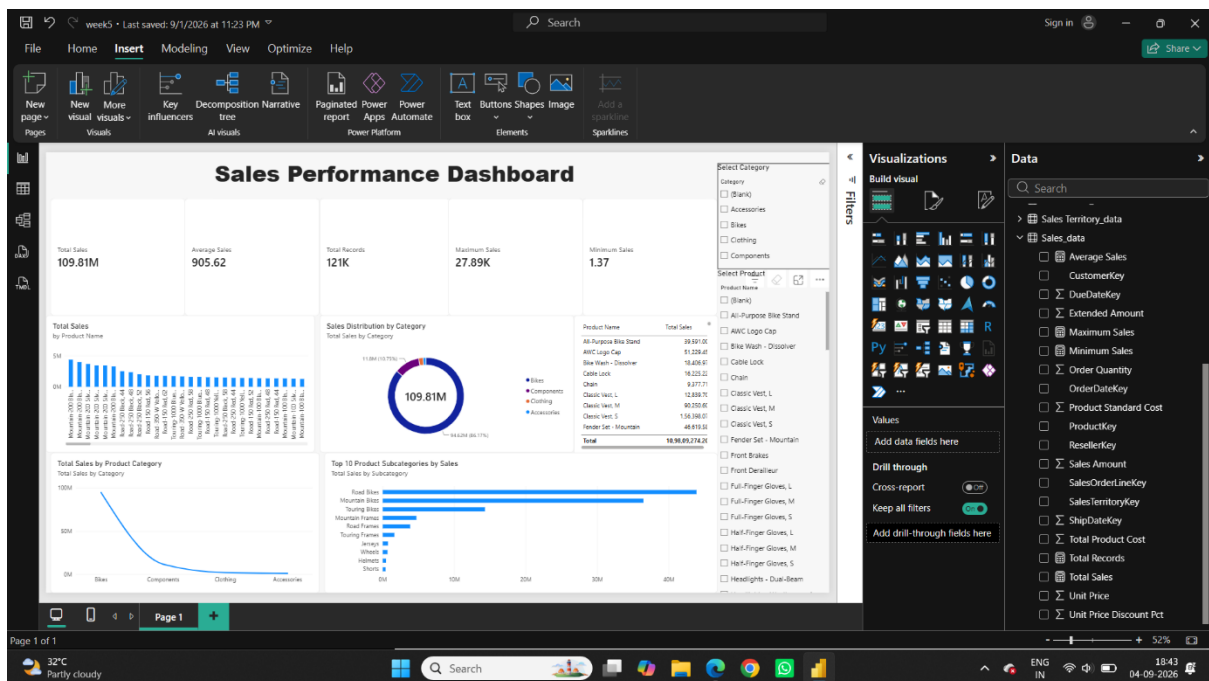
A table visualization was added to display product names along with their corresponding total sales.

This allows individual product performance to be compared easily.

Product Name	Total Sales
All-Purpose Bike Stand	39,591.00
AWC Logo Cap	51,229.45
Bike Wash - Dissolver	18,406.97
Cable Lock	16,225.22
Chain	9,377.71
Classic Vest, L	12,839.70
Classic Vest, M	90,250.60
Classic Vest, S	1,56,398.07
Fender Set - Mountain	46,619.58
Front Brakes	50,299.31
Total	10,98,09,274.20

9) Product Name and Total Sales table

Result



The visualizations successfully transformed the raw sales data into an easy-to-understand graphical representation of business performance.

TASK 6: INTERACTIVE DASHBOARD AND FILTERS

Objective

The objective of this task is to create an interactive Power BI dashboard that allows users to explore sales performance based on different categories and products.

Description

An interactive dashboard was created by combining KPI cards, charts, a product table and slicers into a single report page.

The dashboard was given the title:

Sales Performance Dashboard

The dashboard contains five KPI cards:

1. Total Sales
2. Average Sales
3. Total Records
4. Maximum Sales
5. Minimum Sales

It also contains different charts for analyzing product and category performance.

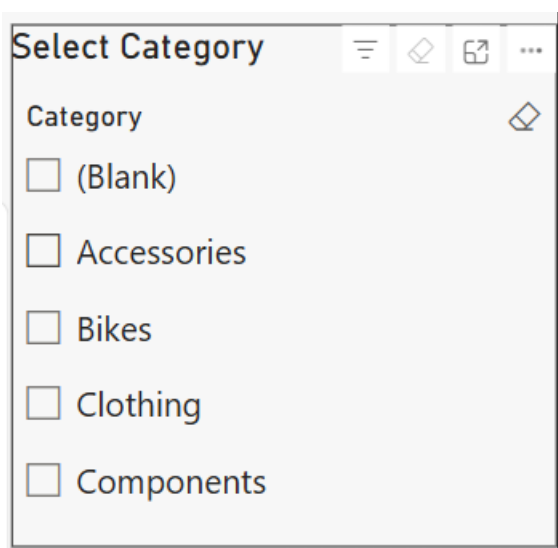
Category Slicer

A **Category slicer** was added to allow users to filter the dashboard according to product category.

The available categories include:

- Accessories
- Bikes
- Clothing
- Components

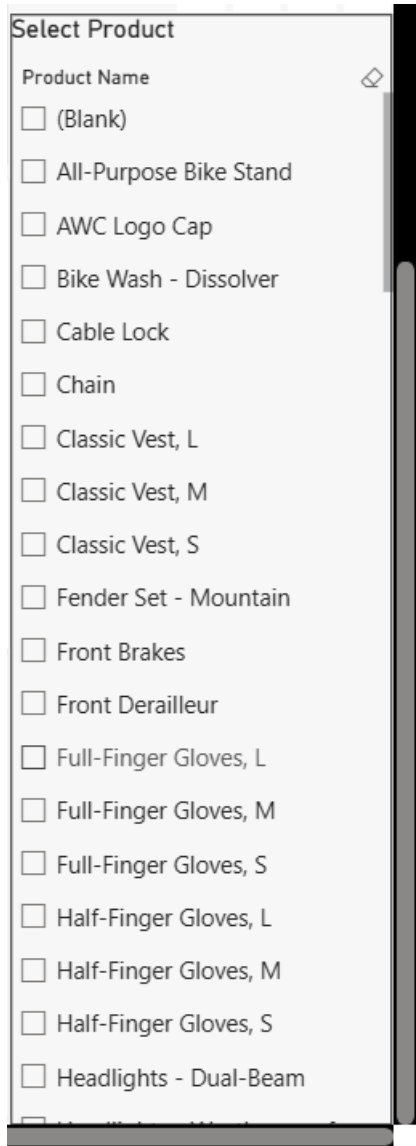
When a category is selected, the connected visuals update according to the selected category.



Product Slicer

A **Product Name slicer** was also added to allow users to select individual products.

This makes it possible to examine the sales performance of a specific product.



Select Product

Product Name

- ☐ (Blank)
- ☐ All-Purpose Bike Stand
- ☐ AWC Logo Cap
- ☐ Bike Wash - Dissolver
- ☐ Cable Lock
- ☐ Chain
- ☐ Classic Vest, L
- ☐ Classic Vest, M
- ☐ Classic Vest, S
- ☐ Fender Set - Mountain
- ☐ Front Brakes
- ☐ Front Derailleur
- ☐ Full-Finger Gloves, L
- ☐ Full-Finger Gloves, M
- ☐ Full-Finger Gloves, S
- ☐ Half-Finger Gloves, L
- ☐ Half-Finger Gloves, M
- ☐ Half-Finger Gloves, S
- ☐ Headlights - Dual-Beam

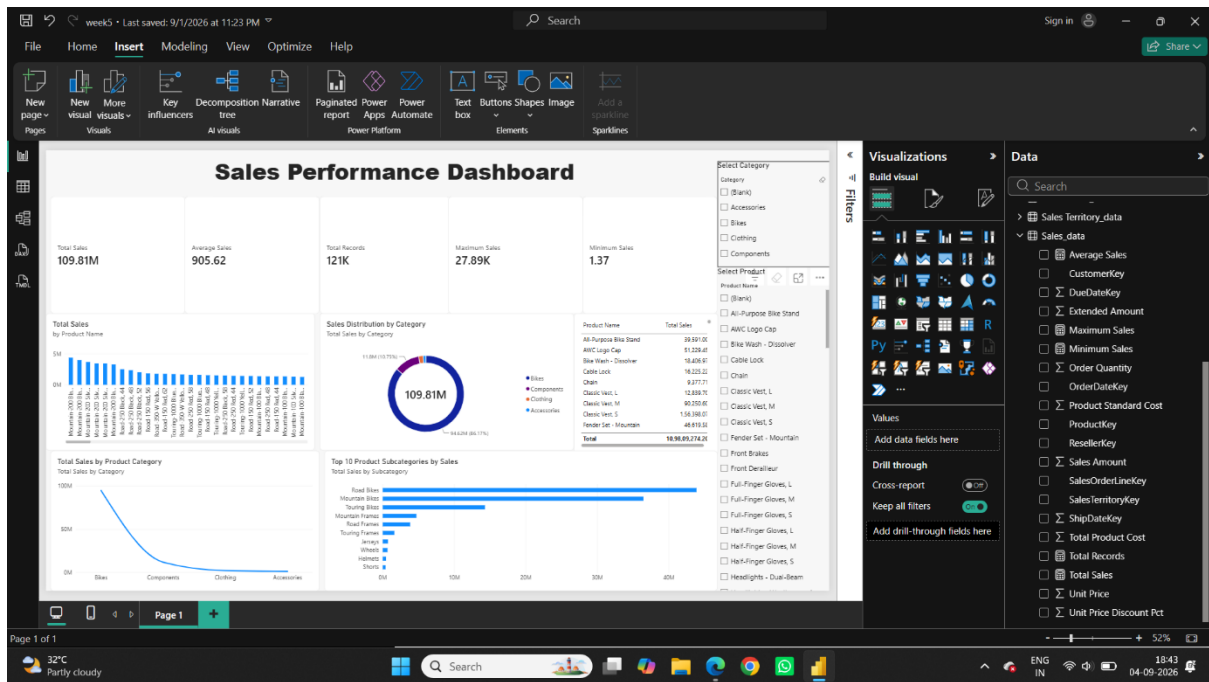
Interactive Features

The dashboard allows users to:

- Filter sales by category.
- Filter sales by product.
- Compare product performance.
- Compare category performance.
- View KPI values.
- Examine top-performing subcategories.
- Analyze the sales distribution across categories.

Result

An interactive sales dashboard was successfully created. The dashboard provides users with a consolidated view of sales performance and allows them to explore the data using interactive filters.



10) Final Sales Performance Dashboard showing all KPI cards, charts, table and slicers.

TASK 7: BUSINESS INSIGHTS AND RECOMMENDATIONS

Objective

The objective of this task is to analyze the Power BI dashboard and identify meaningful business insights, important trends, highest and lowest-performing categories, and recommendations supported by the data.

7.1 Business Insights

Insight 1: Strong Overall Sales Performance

The dashboard shows total sales of approximately **₹109.81 million** from around **121K records**. The average sales value is approximately **₹905.62**.

This indicates that the business has generated substantial sales across its product portfolio.

Insight 2: Bikes Are the Highest-Performing Category

The **Bikes** category is the dominant revenue-generating category.

It contributes approximately **₹94.62 million**, representing around **86.17% of total sales**.

This indicates that customer demand is heavily concentrated on bike products.

Insight 3: Road Bikes Are the Top Subcategory

The **Road Bikes** subcategory is the strongest-performing subcategory, generating approximately **₹43 million** in sales.

This indicates that Road Bikes are an important contributor to the company's overall revenue.

Insight 4: Mountain Bikes Also Perform Strongly

Mountain Bikes generate approximately **₹36 million** in sales and are another major contributor to the business.

The strong performance of both Road Bikes and Mountain Bikes shows that bike-related products are driving the company's sales.

Insight 5: Sales Are Concentrated in a Few Product Groups

The dashboard shows that sales are not evenly distributed among all product categories and subcategories.

A large proportion of revenue comes from Bikes, particularly Road Bikes and Mountain Bikes, while Clothing and Accessories contribute comparatively less.

7.2 Important Trends and Patterns

The following trends were identified from the dashboard:

- The **Bikes category dominates total sales**.
- Road Bikes are the highest-performing subcategory.
- Mountain Bikes are another major contributor to revenue.
- Components, Clothing and Accessories contribute significantly less than Bikes.
- Sales are concentrated among a relatively small number of high-performing products and subcategories.
- Lower-performing categories provide opportunities for future growth.

7.3 Highest and Lowest Performing Categories

Highest Performing Category

Bikes is the highest-performing category, generating approximately **₹94.62 million** in sales.

Lowest Performing Category

Accessories is among the lowest-performing categories and contributes only a small portion of total sales compared with Bikes.

Highest Performing Subcategory

Road Bikes are the highest-performing subcategory with approximately **₹43 million** in sales.

7.4 Business Recommendations

Recommendation 1: Focus on High-Performing Bike Products

The company should prioritize **Road Bikes and Mountain Bikes** in inventory planning and marketing activities because these products generate the highest sales.

Maintaining sufficient stock of these products can help the company meet customer demand and avoid lost sales opportunities.

Recommendation 2: Improve Accessories and Clothing Sales

The company should introduce targeted promotions for lower-performing categories such as **Accessories and Clothing**.

Discounts, seasonal offers and bundle deals can encourage customers to purchase these products along with bikes.

Recommendation 3: Implement Cross-Selling

The company can use cross-selling strategies to increase revenue from lower-performing categories.

For example, when a customer purchases a bike, the business can recommend related products such as:

- Helmets
- Bike locks
- Cycling clothing
- Gloves
- Other accessories

This can increase the value of each customer transaction.

CONCLUSION

The Power BI analysis successfully transformed the sales dataset into an interactive and informative dashboard. The analysis revealed that the **Bikes category is the primary source of revenue**, with Road Bikes and Mountain Bikes being the strongest-performing subcategories.

The dashboard also identified lower-performing categories such as Accessories and Clothing, which provide opportunities for improvement.

Based on these findings, the business should continue investing in high-performing bike products while using **cross-selling, promotional campaigns and bundle offers** to increase sales of lower-performing categories.

Overall, the Power BI dashboard provides useful insights that can support **data-driven business decisions, inventory planning, marketing strategies and revenue growth**.